

Delivery System

ERPNext COURIER INTEGRATION USER GUIDE

Complete Operational & Technical Documentation for Multi-Tenant
ERPNext v15/v16 Courier Integration (Steadfast, Pathao, RedX)

Version 1.0.0 — Production Edition

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1. Executive Summary & Architecture

The **Delivery System** custom app provides seamless, production-ready integration between ERPNext v15/v16 and Bangladeshi courier services (starting with **Steadfast Courier**, with pluggable support for Pathao, RedX, Paperfly, and eCourier).

Key Operational Highlights

- **One-Click & Bulk Booking:** Book orders directly from Sales Order or Delivery Note forms or in bulk (up to 500 per batch) from List Views.
- **Automated Real-Time Tracking:** Automated 30-minute status polling plus inbound webhook listener for instant status updates.
- **Automated Accounting:** Automatic Journal Entries for courier clearing, Payment Entries for payouts, delivery charge splitting, and variance adjustments.
- **8 Script Reports & Dashboard:** Complete suite of delivery, stuck parcel, reconciliation, and financial performance reports.

Module Overview

Component	DocType / File	Description
Courier Master	Courier Provider	Master list of supported couriers (Steadfast, Pathao, RedX, etc.)
Settings	Courier Settings	Global options, source doctype selection, accounting automation toggles
Company Accounts	Courier Account	Child table mapping API keys and GL accounts per ERPNext company
Consignment DocType	Delivery Order	Submittable core tracking record (DS-YYYY-XXXXX) with audit log
Payout Audit Log	Courier Payout Log	Tracks courier payout batches, charges deducted, and Payment Entries

2. Installation & Configuration Guide

2.1 Installation Steps

To install the app on your Frappe bench and target site:

```
# 1. Fetch the repository
bench get-app delivery_system https://github.com/sadikhossainnub/delivery_system.git

# 2. Install on target ERPNext site
bench --site erp.yourdomain.com install-app delivery_system

# 3. Run database migrations
bench --site erp.yourdomain.com migrate
```

2.2 Courier Settings Setup

1. Navigate to **Delivery System > Courier Settings** in Desk.
2. Set **Default Provider** to `Steadfast`.
3. Set **Booking Source DocType**:
 - `Both` : Allow booking from both Sales Order and Delivery Note.
 - `Sales Order` : Restrict booking to Sales Orders only.
 - `Delivery Note` : Restrict booking to Delivery Notes only.
4. Check **Enable Accounting Automation** (enabled by default).
5. Set **Auto Post Variance Threshold** (e.g., `50.0` Taka).

2.3 Configuring Company Credentials & Accounts

In the **Courier Accounts** child table, add a row for each company:

Setting Field	Recommended Value	Description
Company	Your Company Name	Target ERPNext company
Courier Provider	Steadfast	Selected courier service
API Key	your_api_key	From Steadfast merchant panel (portal.packzy.com)
Secret Key		Encrypted password field (never exposed in logs/UI)
Clearing Account	Steadfast Clearing Account	Asset (Bank/Current Asset) holding account
Delivery Charge Account	Delivery Charges	Expense account for shipping fees
Variance Account	Courier Variance	Expense/Income account for small COD rounding differences
	Steadfast COD	Mode of Payment record linked to Bank Account

Setting Field	Recommended Value	Description
Default Mode of Payment		

3. Parcel Booking & Tracking Operations

3.1 Single Parcel Booking

1. Open a submitted **Sales Order** or **Delivery Note**.
2. Click the **Send to Courier** button in the top action bar.
3. Review recipient details in the modal dialog:
 - **Recipient Phone:** Must be a valid 11-digit Bangladeshi mobile number (e.g., 01712345678).
 - **Recipient Address:** Automatically truncated to 250 characters max.
 - **COD Amount:** Defaults to document grand total.
 - **Delivery Type:** Home Delivery (HD) or Point Delivery (PD).
4. Click **Book Courier**. The system auto-creates and submits a `Delivery Order` , registers the consignment with Steadfast, and populates the `Consignment ID` and `Tracking Code` .

3.2 Bulk Booking from List View

1. Go to **Sales Order List** or **Delivery Note List**.
2. Select multiple submitted orders via checkboxes.
3. Click **Actions > Send Selected to Courier**.
4. The app processes orders in batch (up to 500 items) and displays a detailed success/failure summary.

3.3 Status Lifecycle & Badges

Status	Badge Color	Meaning
<code>pending</code>	Orange	Consignment booked with courier, pending pickup
<code>in_review</code>	Yellow / Orange	Parcel in transit / under review at courier hub
<code>delivered</code>	Green	Parcel successfully delivered to recipient
<code>cancelled</code>	Red	Parcel cancelled or returned to merchant
<code>hold</code>	Grey	Delivery placed on hold by courier

3.4 Inbound Webhooks & Status Polling

- **Background Scheduler:** `sync_pending_deliveries` runs every 30 minutes to refresh non-terminal orders.
- **Inbound Webhook Endpoint:** Configure `https://your-site/api/method/delivery_system.webhook.steadfast_webhook` in Steadfast merchant panel for real-time push updates.

4. Automated Accounting & Reconciliation

4.1 Automated Journal & Payment Entries Flow

The app automates all GL postings using standard ERPNext accounting APIs without requiring manual bookkeeping:

Phase 1: Delivery Confirmation (Clearing Entry)

When an order is marked **delivered**, the system posts a Journal Entry:

Debit: Steadfast Clearing Account (Asset) [COD Amount]

Credit: Debtors (Customer Receivable) [COD Amount]

Effect: Transfers receivable from customer to courier clearing account.

Phase 2: Payout & Delivery Charge Posting

When Steadfast releases a payout batch:

- **Net Payout:** Payment Entry (Debit: Bank Account / Credit: Clearing Account)

- **Delivery Charges:** Journal Entry (Debit: Delivery Charges Expense / Credit: Clearing Account)

Effect: Clearing account nets to zero for all reconciled consignments.

Phase 3: Variance & Cancellation Reversals

- **Small Variance (≤ 50 Taka):** Auto-posted to Courier Variance account.

- **Large Variance (> 50 Taka):** Flagged in COD Reconciliation report for manual review.

- **Order Cancellation Post-Delivery:** Automatically posts an equal-and-opposite reversal Journal Entry.

5. Reports & Analytics Suite

The app includes 8 comprehensive Script Reports accessible from Desk:

Report Name	Key Purpose	Filters
1. Courier Delivery Status	Single view of all delivery orders, consignment IDs, status, and days pending	Date range, Provider, Status, Company
2. Pending Stuck Deliveries	Surfaces parcels stuck $\geq$ N days (default 3) for operational follow-up	Min Days Pending, Provider, Company
3. COD Reconciliation	Matches expected COD vs courier payouts. Includes "Mark Manually Reconciled" button	Date range, Reconciled Status, Company
4. Return Cancelled Orders	Analyzes cancelled/returned orders with log reasons for return pattern analysis	Date range, Provider, Company
5. Courier Performance	Multi-courier comparison: Total Bookings, Delivered Count, Success Rate %, Avg Time	Date range, Company
6. Zone Wise Delivery	Delivery success rate grouped by geographic district/area	Date range, Provider, Company
7. Delivery Charge vs Revenue	Per-order margin analysis (Invoice Amount – Delivery Charges – COD Fees)	Date range, Company
8. Courier Clearing Statement	Mini ledger statement of Clearing Account: Opening Balance, Dr/Cr, Closing Balance	Date range, Company

5.1 Dashboard Number Cards

- **Courier Account Balance:** Live balance from Steadfast API (cached 15 minutes).
- **Today's Bookings:** Total consignments booked today.
- **Today Delivered:** Total orders marked delivered today.
- **Today Cancelled:** Total orders cancelled today.

6. Developer & Extensibility Guide

6.1 Architecture Overview

```
delivery_system/
├─ couriers/
│   ├── __init__.py      # BaseCourierClient ABC + get_client() factory
│   ├── steadfast.py     # Full Steadfast implementation (12 API endpoints)
│   ├── pathao.py        # Pluggable stub
│   └── redx.py           # Pluggable stub
├─ api.py                # Whitelisted endpoints
├─ accounting.py         # Automated GL / JE / PE posting engine
├─ tasks.py              # Scheduled background jobs
└─ webhook.py            # Inbound webhook processing
```

6.2 Adding a New Courier Service

To integrate a new courier (e.g., Pathao or RedX):

1. Create `delivery_system/couriers/mycourier.py` implementing `BaseCourierClient`.
2. Register the module mapping in `REGISTRY` inside `delivery_system/couriers/__init__.py`:

```
REGISTRY = {
    "steadfast": "delivery_system.couriers.steadfast",
    "mycourier": "delivery_system.couriers.mycourier",
}
```

3. Add a fixture record in `Courier Provider` with `provider_code = "mycourier"`. No core code refactoring required!

6.3 Running Unit Tests

Run the full test suite (14 unit tests) using bench python environment:

```
/home/sayed/frappe-bench/env/bin/python -m unittest discover -s delivery_system/tests -p "test_*.py"
```

Support & Maintenance

For issues, feature requests, or enterprise support, contact **PrimeTechBD** or visit the GitHub repository:
https://github.com/sadikhossainnub/delivery_system